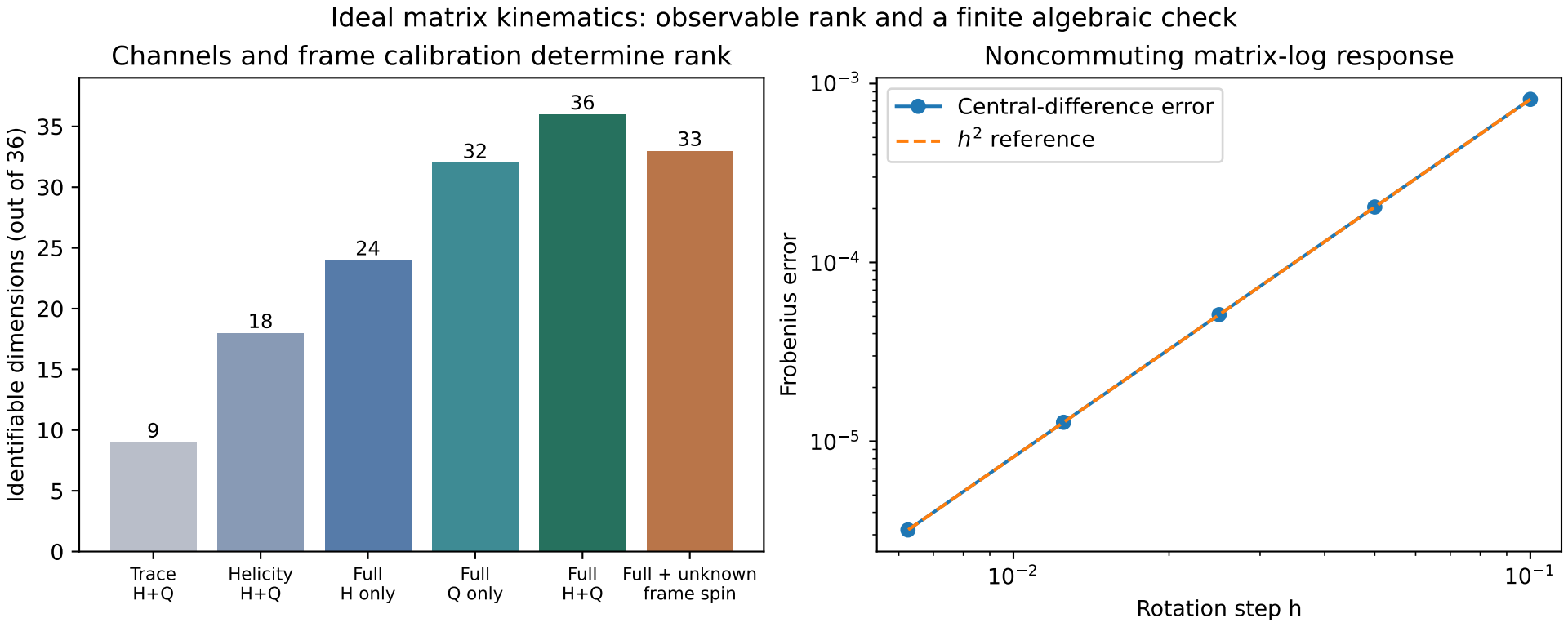
Ideal matrix kinematics: observable rank and a finite algebraic check
Channels and frame calibration determine rank
Identifiable dimensions (out of 36)
0
5
10
15
20
25
30
35
9
18
24
32
36
33
Trace H+Q
Helicity H+Q
Full H only
Full Q only
Full H+Q
Full + unknown frame spin
Noncommuting matrix-log response
Frobenius error
10^{-3}
10^{-4}
10^{-5}
Central-difference error
h^2 reference
10^{-2}
10^{-1}
Rotation step h